

Fakhreddin (Amir) Amirhosseini

Mechatronics Engineer

(672)336-2686 • Vancouver • fakhredin0098@gmail.com • [LinkedIn](#) • [Portfolio](#)

PROFESSIONAL SUMMARY

Mechatronics engineer with nearly a decade of experience taking products from concept to mass production. Hands-on background in mechanical mechanism design, custom part design and iterative prototyping with special materials. Experienced with electronics and software that makes cross-functional collaboration straightforward.

WORK EXPERIENCE

Automation Engineer Intern

Jan 2026 – Present

Kardium– *Burnaby, BC*

- Currently building and maintaining software and hardware for automation jigs used in catheter manufacturing, working within the Windchill PLM environment and Bitbucket.
- Developed and contributed to C# and Python codebases, including UIs, machine vision, hardware control, and data analyzers, following SOLID principles with comprehensive unit testing.
- Contributed to design revision cycles for code, PCB, and technical documents. Made tools for and collaborated with the mechanical, manufacturing, and quality engineering teams.

Mitacs Intern and Graduate Research Assistant

Sep 2023 – Jul 2025

Aluminate Inc. and Intelligent Sensing Lab at Simon Fraser University (SFU) – *Surrey, BC*

- Designed housings and assemblies for sensors use in aluminum smelting environments (1000°C, corrosive molten cryolite), requiring material investigation including ceramics (Al_2O_3 , BN, Si_3N_4 , SiC, graphite) and metals (stainless steel, molybdenum, etc.) to meet structural and chemical requirements.
- Designed parts for CNC machining and coordinated with machinists, ceramic manufacturers, and mechanical part vendors in Canada, the US, and China to fabricate prototypes; managed vendor lead times, pricing negotiations, and delivery schedules to meet R&D milestones.
- Designed 3 novel sensing techniques (resonant capacitive and eddy-current) validated through field testing and COMSOL simulation. Results published in a provisional US patent, an award-winning IEEE Sensors conference paper, and a featured IEEE Sensors Journal article.
- Conducted a material corrosion study by submerging ceramic samples into molten cryolite (a highly corrosive chemical) and comparing thickness and surface finish to identify a suitable coating material
- Organized field test logistics including coordination with a team of 8 engineers and technicians, developing contingency plans, and conducting on-site measurements at a test facility. Used network analyzers for field measurement and COMSOL to model, characterize, and validate sensor behavior.

Product Development Lead

Jun 2017 – Sep 2023

Tart Robotics – Toronto, ON

- Helped take 3 robotic platforms from concept to mass-produced product, with over 3,000 units sold globally by designing the concepts, prototyping, and hardware/software development. [See products](#).
- Designed plastic PCB enclosures, motor housings, shafts, couplings, and snap-fit structural components compatible with Lego Technic parts in SolidWorks, iterating through many 3D-printed prototypes, and applied DFM principles to simplify mold geometry to reduce tooling costs.
- Designed complex mechanical mechanisms including cable-driven systems, multi-bar linkages, and leg locomotion mechanisms. Made exploded views of the projects and assembly files and assembly instructions using SolidWorks and SolidWorks Composer. [See Projects](#).
- Designed proprietary electronics for an in-house low-power ultrasonic sensor and a leak-free soft power switch for circuits with non-removable batteries, resolving manufacturing and stocking challenges.
- Mentored and supported 6 interns during internship programs, providing exposure to mechanical design, software, hardware, and firmware development, resulting in multiple recruitments for the company.

Research Assistant

Sep 2020 – Jul 2023

Kharazmi University – Tehran, Iran

- Researched motion planning for autonomous mobile robots, resulting in LPM-Net: a deep learning model that replicates MPC path planning at 50% less compute, open-sourced for ROS. [Journal](#), [GitHub](#).
- Built machine vision pipelines using YOLO and SSD models with MobileNet frameworks for ROS, implemented on Raspberry Pi for object tracking. [GitHub](#).

Research Assistant

Sep 2018 – Sep 2020

Microfluidics Lab at Qazvin International University – Qazvin, Iran

- Designed and fabricated a microfluidic lab-on-a-chip for circulating tumor cell separation with photolithography and PDMS microfabrication.

Research Assistant

Sep 2017 – Sep 2018

IoT Lab at Qazvin International University – Qazvin, Iran

- Designed a gate-level address-decoder for ultra-low-power self-sufficient IoT devices, significantly reducing power from $\sim 100 \mu W$ to 68 nW, resulting in a journal publication. [Journal link](#).

EDUCATION

Simon Fraser University – Surrey, BC

Sep 2023 – Jan 2026

M.A.Sc. – Mechatronic Systems Engineering

Simon Fraser University Beedie School of Business – Burnaby, BC

Sep 2024 – Apr 2025

Graduate Certificate – i2I, Science and Technology Commercialization

Kharazmi University – Tehran, Iran

Sep 2020 – Jun 2023

M.Sc. – Artificial Intelligence

Qazvin International University - Qazvin, Iran

Sep 2016 – Jun 2020

B.Sc. – Electrical and Control Engineering

AWARDS AND RECOGNITIONS

Nominated for the Governor General's Academic Medal.	Jan 2026
Received Outstanding Lecture Award at IEEE Sensors conference among 400+ accepted works.	Oct 2025
Graduate Fellowship recipient (3x)	2023 - 2025
Recipient of full scholarship for all education levels.	2016 - 2026
(Fun fact!) Featured multiple times on Hackster.com and Instructable.com .	2017 - 2023

MECHANICAL DESIGN PROJECTS

- **Lego Mechanical Design Projects:** Designed a tower crane replica with working lift, slew, and trolley, plus biped and quadruped walking robots each driven by a single DC motor. All mechanisms use custom multi-bar linkages, LEGO Technic, and custom 3D printed parts. [See projects.](#)
- **LEGO-Compatible Mechanical Parts:** Designed and mass-manufactured a full range of custom parts, including PCB enclosures, motor housings, shafts, and couplings; all LEGO Technic-compatible and injection-molded modeled in SolidWorks with tight tolerances and DFM constraints. [Photo gallery.](#)
- **Aluminum Smelting Level Sensor:** Designed a contactless sensor to monitor molten aluminum levels and bath composition in real time inside industrial smelting pots with integrated embedded electronics inside ceramic protective housing, solving a long-standing industry problem. [Watch presentation](#)

VOLUNTEER WORK

- Conference Facilitator at IEEE Sensors 2025, Vancouver
- Sessional robotics instructor for K-12 children.

SKILL

- SolidWorks (part design, assemblies, DFM for injection molding and CNC), rapid prototyping (3D printing), mechanism design, material selection for extreme environments, PLM (Windchill), engineering documentation and design review processes.
- C#, .Net, C/C++, Python, JavaScript, Git. Professional codebases following SOLID principles with comprehensive unit testing. Altium Designer, PCB design for high-frequency and high-voltage applications.
- FEA with COMSOL. R&D problem solving and experimental validation.
- Deep Learning, TensorFlow, machine vision. ROS, robot manipulation, and motion planning.